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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/629,032	01/06/2020	Dale Ehnes	109167-0622	9133

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EXAMINER	
LOPEZ, LESLIE ANN	

ART UNIT	PAPER NUMBER
3774	

NOTIFICATION DATE	DELIVERY MODE
06/14/2021	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

ipdocketing@foley.com

Office Action Summary	Application No. 16/629,032	Applicant(s) Ehnes et al.	
	Examiner LESLIE A LOPEZ	Art Unit 3774	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 4/24/2020.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-2,4,10-11,15-16,18-19,21,27-28,33,36-38,41-42,44 and 46 is/are pending in the application.

5a) Of the above claim(s) 18-19,21,27-28,33,36-38,41-42,44 and 46 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-2,4,10-11 and 15-16 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 1/6/2020 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. The first inventor to file provisions of the Leahy-Smith America Invents Act (AIA) apply to any application for patent, and to any patent issuing thereon, that contains or contained at any time—

(A) a claim to a claimed invention that has an effective filing date on or after March 16, 2013 wherein the effective filing date is:

(i) if subparagraph (ii) does not apply, the actual filing date of the patent or the application for the patent containing a claim to the invention; or

(ii) the filing date of the earliest application for which the patent or application is entitled, as to such invention, to a right of priority under 35 U.S.C. 119, 365(a), or 365(b) or to the benefit of an earlier filing date under 35 U.S.C. 120, 121, or 365(c); or

(B) a specific reference under 35 U.S.C. 120 , 121, or 365(c), to any patent or application that contains or contained at any time a claim as defined in paragraph (A), above.

Status of the Claims

3. Claim(s) 1-2, 4, 10-11, 15-16, 18-19, 21, 27-28, 33, 36-38, 41-42, 44, and 46 is/are pending. Claim(s) 18-19, 21, 27-28, 33, 36-38, 41-42, 44, and 46 is/are

withdrawn. Claim(s) 3, 5-9, 12-14, 17, 20, 22-26, 29-32, 34-35, 39-40, 43, 45, and 47 is/are canceled.

Election/Restriction

4. REQUIREMENT FOR UNITY OF INVENTION

As provided in 37 CFR 1.475(a), a national stage application shall relate to one invention only or to a group of inventions so linked as to form a single general inventive concept ("requirement of unity of invention"). Where a group of inventions is claimed in a national stage application, the requirement of unity of invention shall be fulfilled only when there is a technical relationship among those inventions involving one or more of the same or corresponding special technical features. The expression "special technical features" shall mean those technical features that define a contribution which each of the claimed inventions, considered as a whole, makes over the prior art.

The determination whether a group of inventions is so linked as to form a single general inventive concept shall be made without regard to whether the inventions are claimed in separate claims or as alternatives within a single claim. See 37 CFR 1.475(e).

When Claims Are Directed to Multiple Categories of Inventions:

As provided in 37 CFR 1.475 (b), a national stage application containing claims to different categories of invention will be considered to have unity of invention if the claims are drawn only to one of the following combinations of categories:

(1) A product and a process specially adapted for the manufacture of said product; or

(2) A product and a process of use of said product; or

(3) A product, a process specially adapted for the manufacture of the said product, and a use of the said product; or

(4) A process and an apparatus or means specifically designed for carrying out the said process; or

(5) A product, a process specially adapted for the manufacture of the said product, and an apparatus or means specifically designed for carrying out the said process.

Otherwise, unity of invention might not be present. See 37 CFR 1.475 (c).

5. This application contains claims directed to more than one species of the generic invention. These species are deemed to lack unity of invention because they are not so linked as to form a single general inventive concept under PCT Rule 13.1.

The species are as follows:

Species of <u>overall device design</u>

Species 1: Figure 3B, e.g. claims 1-2, 4, 10-11, 16

Species 2: Figures 7A-D, e.g. claims 18-19, 21, 27-28

Species 3: Figure 9D, e.g. claims 33, 36-37, 41

Species 4: Figure 10A, e.g. claims 33, 36, 38,

Species 5: Figure 10B, e.g. claims 33, 36-37, 41

Species 6: Figure 12A, 13A, e.g. claims 33, 36-37, 41-42, 44

Species 7: Figure 15C, e.g. claims 33, 36-37, 41-42, 46

Species 8: Figure 16C, e.g. claims 33, 36, 46

Applicant is required, in reply to this action, to elect a single species to which the claims shall be restricted if no generic claim is finally held to be allowable. The reply must also identify the claims readable on the elected species, including any claims subsequently added. An argument that a claim is allowable or that all claims are generic is considered non-responsive unless accompanied by an election.

Upon the allowance of a generic claim, applicant will be entitled to consideration of claims to additional species which are written in dependent form or otherwise require all the limitations of an allowed generic claim. Currently, the following claim(s) are generic: none.

6. The groups of inventions listed above do not relate to a single general inventive concept under PCT Rule 13.1 because, under PCT Rule 13.2, they lack the same or corresponding special technical features for the following reasons:

The Species supra lack unity of invention because even though the inventions of these groups require the technical feature of an endovascular system having a main tubular graft body deployable in a main artery, this technical feature is not a special technical feature as it does not make a contribution over the prior art in view of Arbefeuille, et al (Arbefeuille) (US 2013/0184806) teaches an endovascular system (e.g. abstract, Figures 23A-C) having a main tubular graft body deployable in a main artery (e.g. Figures 23 A-C, [0072]).

7. Applicant is reminded that upon the cancellation of claims to a non-elected invention, the inventorship must be corrected in compliance with 37 CFR 1.48(a) if one or more of the currently named inventors is no longer an inventor of at least one claim remaining in the application. A request to correct inventorship under 37 CFR 1.48(a) must be accompanied by an application data sheet in accordance with 37 CFR 1.76 that identifies each inventor by his or her legal name and by the processing fee required under 37 CFR 1.17(i).

8. During a telephone conversation with Victor de Gyarfas on 3/12/2021 a provisional election was made with traverse to prosecute the invention of Species 1 (Figure 3B), claims 1-2, 4, 10-11, and 15-16. Affirmation of this election must be made by applicant in replying to this Office action. Claims 18-19, 21, 27-28, 33, 36-38, 41-42, 44, 46 are withdrawn from further consideration by the examiner, 37 CFR 1.142(b), as being drawn to a non-elected invention.

Claim Objections

9. Claim 1 is objected to because of the following informalities:

10. Claim 1 recites “stent segment are”, which should be “segment is” or “segments are”.

11. Appropriate correction is required.

Claim Rejections - 35 USC § 112

12. The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

13. Claim 15 is rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

14. **Claim 15** recites "one or more adjacent stent segments". It is unclear if these segments are part of the "at least one stent segment" introduced in claim 1 or separate segment(s).

Claim Rejections - 35 USC § 102

15. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

16. Claim(s) 1-2, 4, 10-11, and 15-16 is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Vinluan (US 2013/0103135 A1).

Regarding **Claim 1**, Vinluan teaches **an endovascular system** (e.g. Figures 1, 21-22) **comprising:**

a main tubular graft body (e.g. Figure 1) ***configured to be deployable within a main artery*** *(e.g. Figure 1)*, **wherein the main tubular graft body comprises a tubular graft wall** (there is inherently a wall preset);

a plurality of inflatable channels disposed along the main tubular graft body (e.g. Figures 1-2, #s 56)**; and**

at least one stent segment disposed along the tubular graft wall of the main tubular graft body (e.g. Figures 1, 4, #30b)**,**

wherein the plurality of inflatable channels *are configured to be inflated with an inflation medium* *(e.g. [0058], the channels are inflatable and therefore must use some kind of medium)***, and**

wherein one or more of the at least one stent segment are disposed between two or more adjacent inflatable channels of the plurality of inflatable channels (e.g. stent #30b)**.**

Regarding **Claim 2**, the at least one stent is at least one ring segment (e.g. Figure 4, [0059], #30b).

Regarding **Claim 4**, the at least one stent is a continuous wire wound stent (e.g. Figure 4, [0059], #30b).

Regarding **Claim 10**, the graft body has at least one fenestration disposed between two or more adjacent inflatable channels of the plurality of inflatable channels (e.g. Figures 1-2, #26) and *configured to be in fluid contact with at least one branch artery branched from the main artery (e.g. [0043], [0010]).*

Regarding **Claim 11**, the fenestration is covered by a stent mesh (e.g. Figure 21, mesh formed by the ring stents of the extension branch).

Regarding **Claim 15**, the fenestration is bounded by two or more adjacent inflatable channels and one or more stent segments (e.g. Figures 1-2, 4).

Regarding **Claim 16**, there is an anchoring member attached to a proximal end, a distal end or both ends of the graft body (e.g. Figure 1, z-stents at axial ends).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LESLIE ANN LOPEZ whose telephone number is (571)270-7044. The examiner can normally be reached on 8:30 AM - 5:30 PM, MST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jerrah Edwards, can be reached at 408-918-7557. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Leslie Lopez/
Primary Examiner, Art Unit 3774
6/8/2021